

Database System Labs

- Prepare

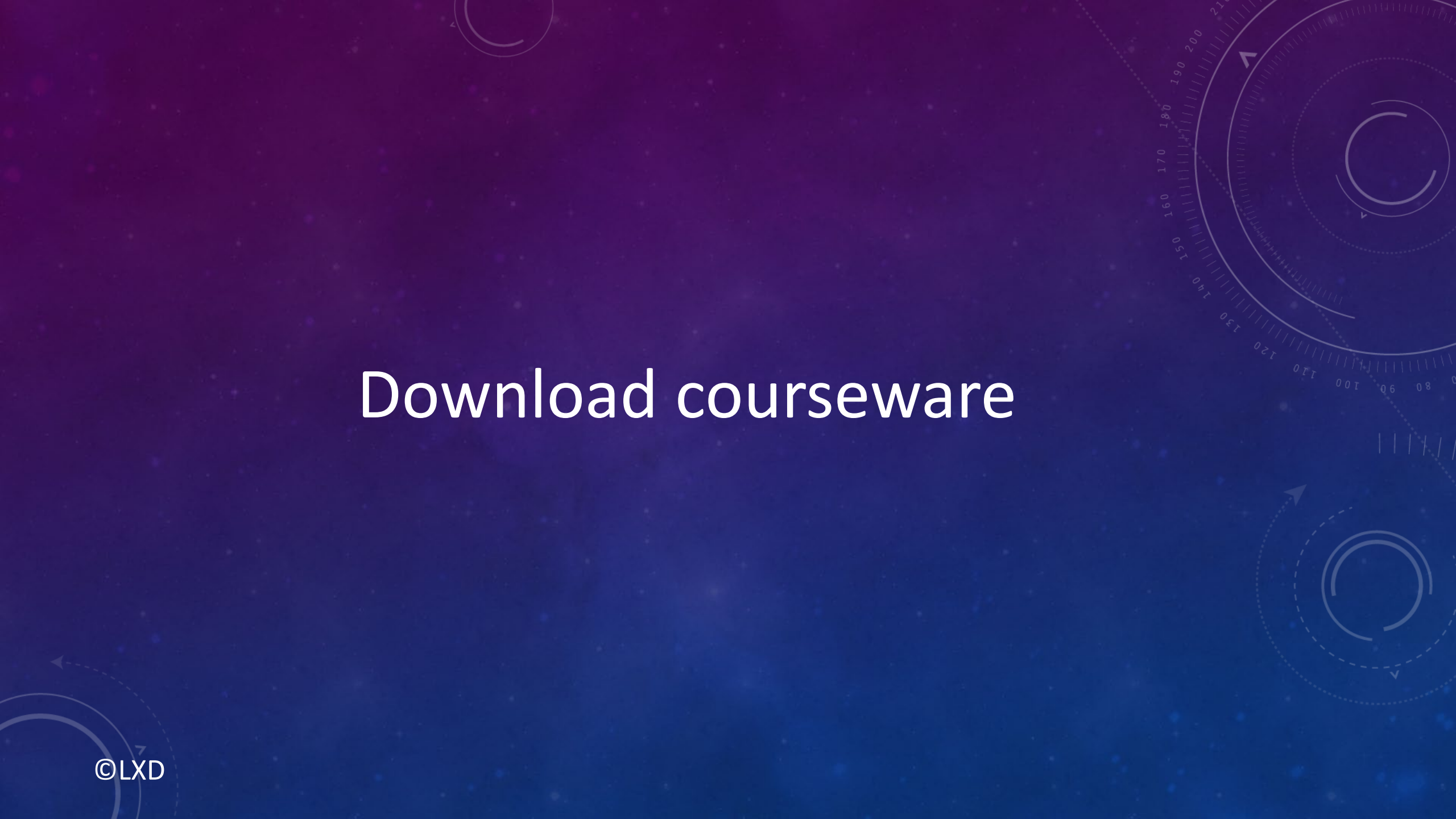
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Nankai University

OBJECTIVES

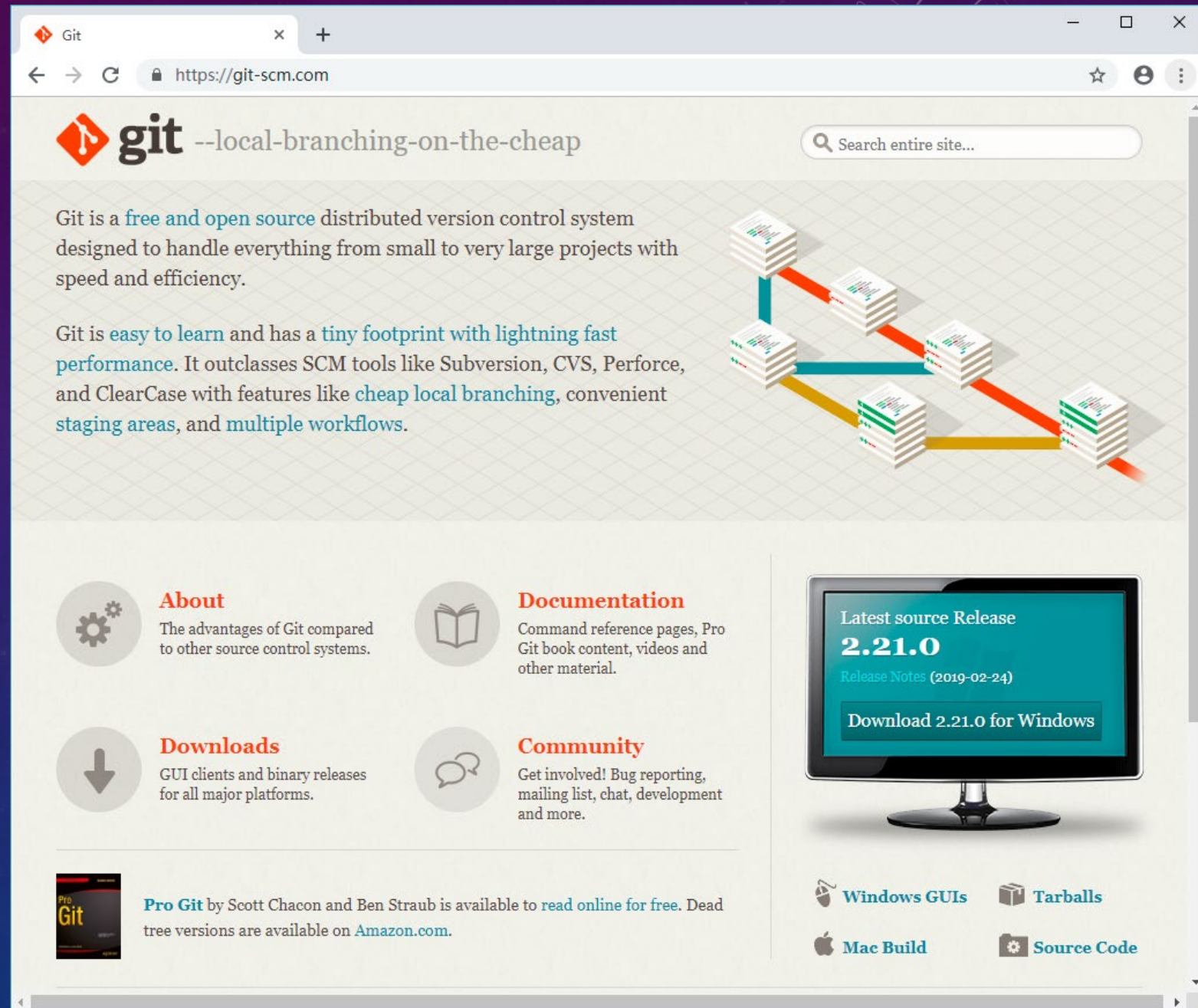
- Download courseware
- DBMS Installation: Server (MySQL), Client (MysqlWorkbench)
- DBMS Login using super user
- Install Python3.13 and Pycharm
- Connect MySQL using python

The background is a gradient of dark blue and purple, speckled with small white dots resembling stars. On the right side, there are faint, white circular patterns that look like technical diagrams or orbits, with some numerical markings (e.g., 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210) and arrows. In the bottom left corner, there is a small circular graphic with a dashed line and an arrow.

Download courseware

LABS PREPARE

- git
 - <https://git-scm.com>
 - Download
 - Install



git clone https://github.com/albertleecn/dblabs.git

LABS PREPARE

- material of DB course
 - git clone
<https://github.com/albertleecn/dblabs>
 - git pull
- Get the latest documentation

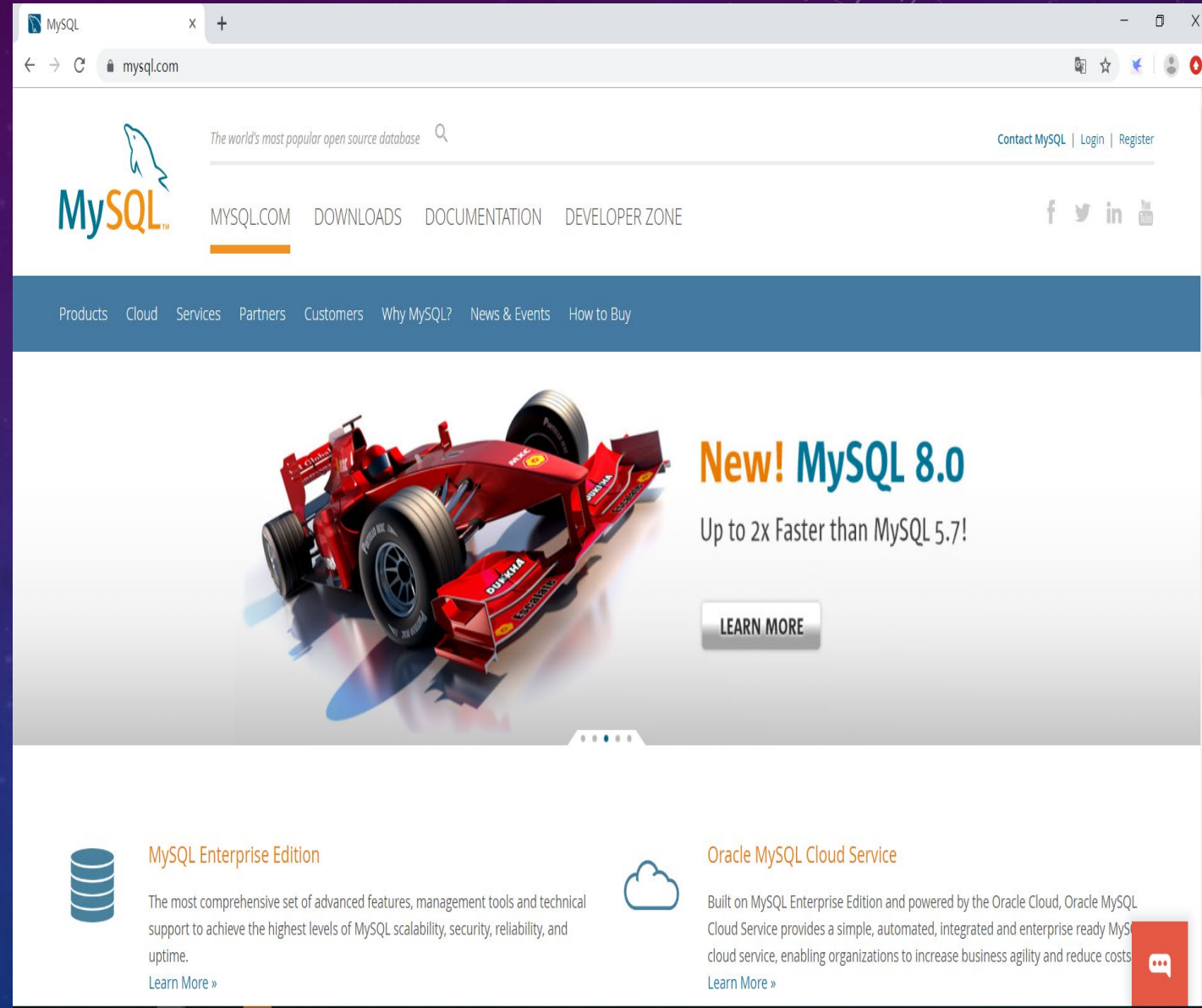
The screenshot shows the GitHub repository page for 'albertleecn/dblabs'. The page includes a navigation bar with links to 'Why GitHub?', 'Enterprise', 'Explore', 'Marketplace', and 'Pricing'. Below the navigation bar, the repository name 'albertleecn / dblabs' is displayed, along with 'Watch', 'Star', and 'Fork' buttons. The 'Code' tab is selected, showing a list of files and their commit history. The files listed are 'labs', 'papers', 'README.md', 'lecture01DbSystemIntroR6.pdf', 'lecture02DbSystemRDbModelR6.pdf', and 'lecture03DbSystemSQL1basicR6.pdf'. The commit history for each file is shown, including the commit message and the time since the last commit.

File	Commit Message	Time
labs	initial labs	11 minutes ago
papers	ok	18 minutes ago
README.md	modify README.md	a year ago
lecture01DbSystemIntroR6.pdf	add new lecture01	a year ago
lecture02DbSystemRDbModelR6.pdf	modify lecture02	a day ago
lecture03DbSystemSQL1basicR6.pdf	add lecture03	11 months ago

Install mysql db server and client

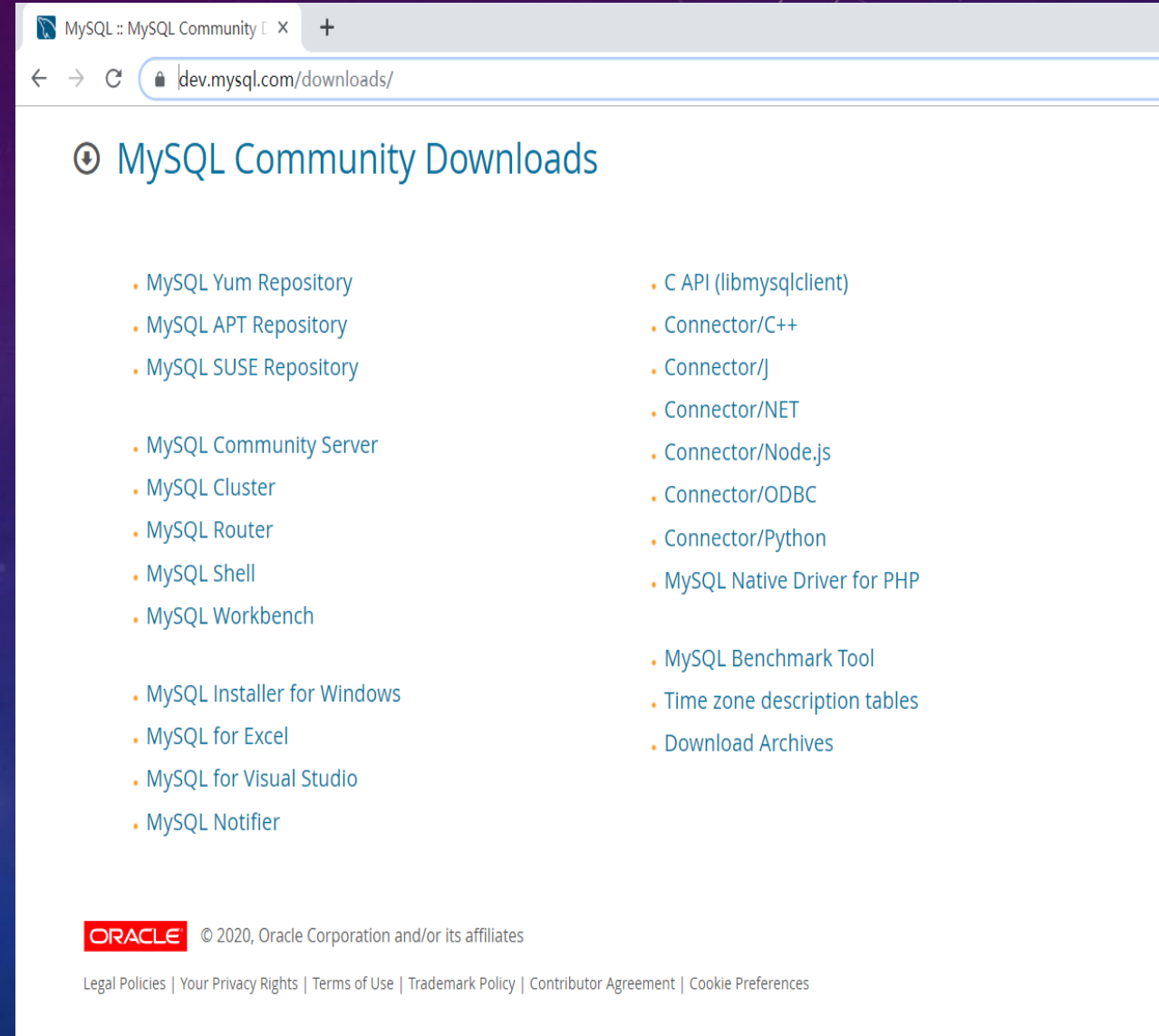
LABS PREPARE

- mysql
 - <https://www.mysql.com/>
 - MySQL 8.0
 - MySQL Community Edition (GPL)



LABS PREPARE

- mysql
 - <https://www.mysql.com/>
 - MySQL 8.0
 - MySQL Community Edition (GPL)
 - Server & Client & Connectors
 - **MySQL Installer for Windows**



DOWNLOADS

<https://dev.mysql.com/downloads/installer/>

The screenshot shows the MySQL Community Downloads page for the MySQL Installer. The browser address bar shows the URL `dev.mysql.com/downloads/installer/`. The page title is "MySQL Community Downloads" with a sub-header "MySQL Installer". There are two tabs: "General Availability (GA) Releases" (selected) and "Archives".

MySQL Installer 8.0.36

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.36

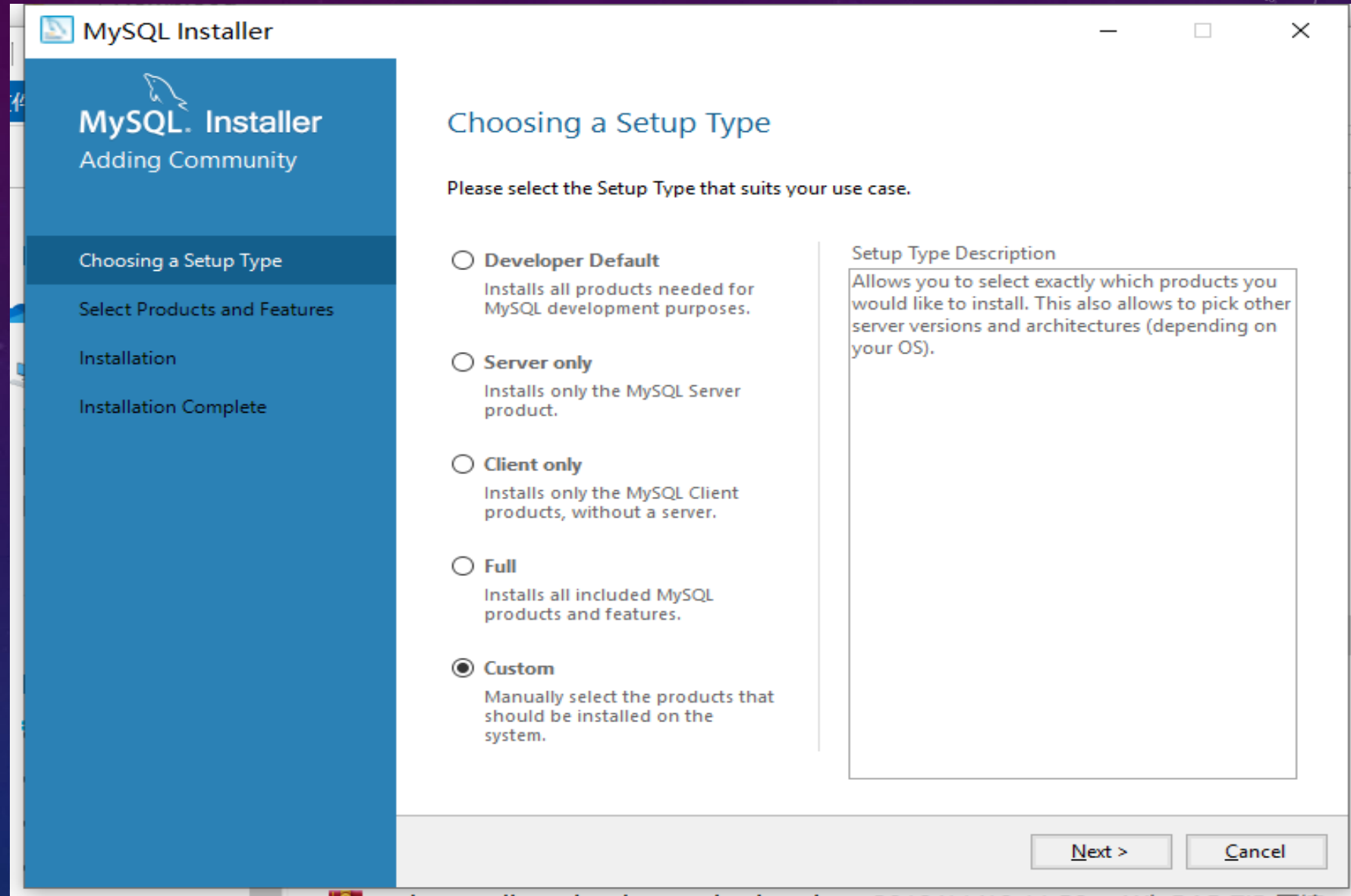
Select Operating System:
Microsoft Windows

Operating System	Version	Size	Download
Windows (x86, 32-bit), MSI Installer	8.0.36	2.1M	Download
(mysql-installer-web-community-8.0.36.0.msi)			
MD5: 81061532541f716cf6c6e2c4881a154c Signature			
Windows (x86, 32-bit), MSI Installer	8.0.36	285.3M	Download
(mysql-installer-community-8.0.36.0.msi)			
MD5: d63232c190d0c9c294a2f8d776ed1c20 Signature			

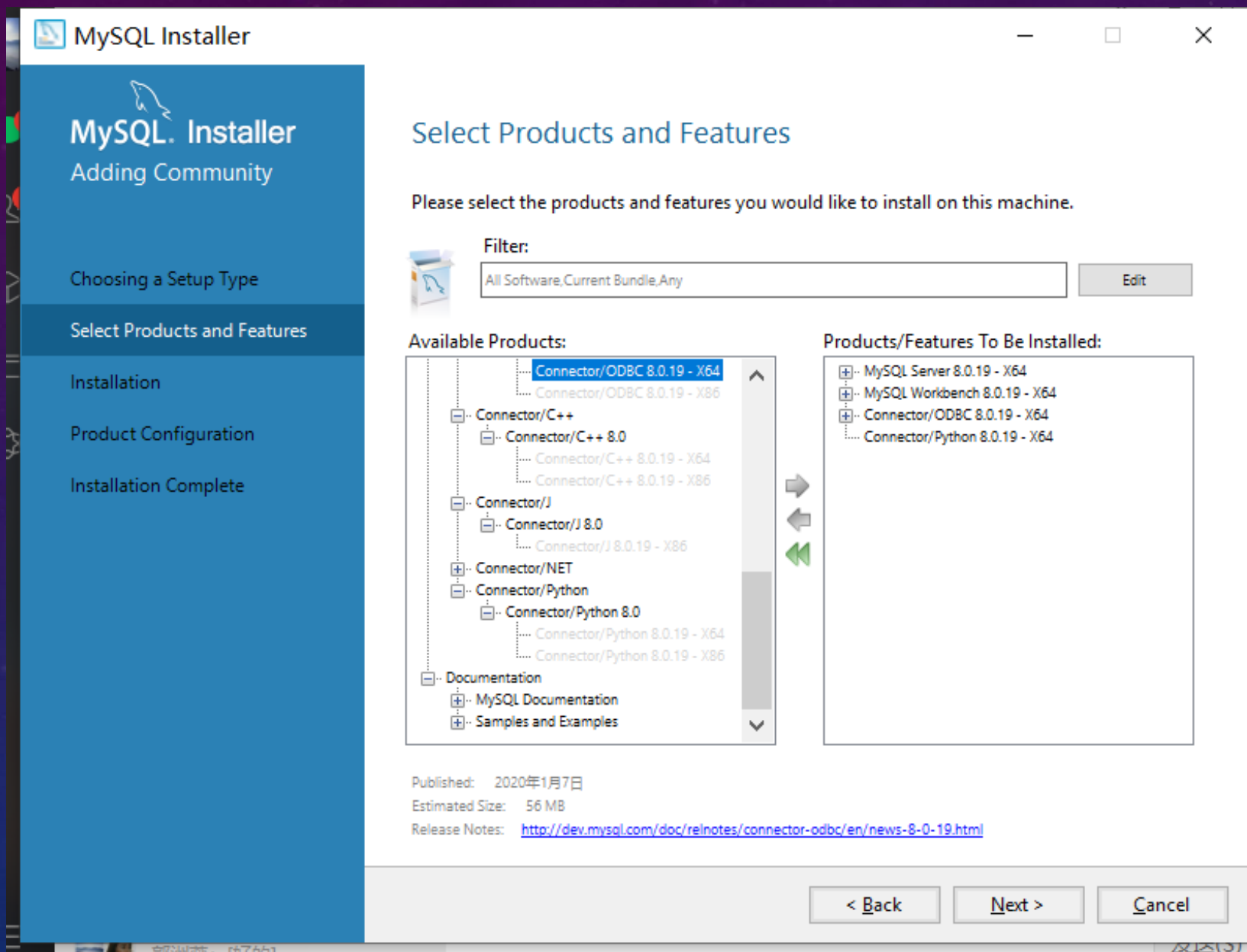
We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

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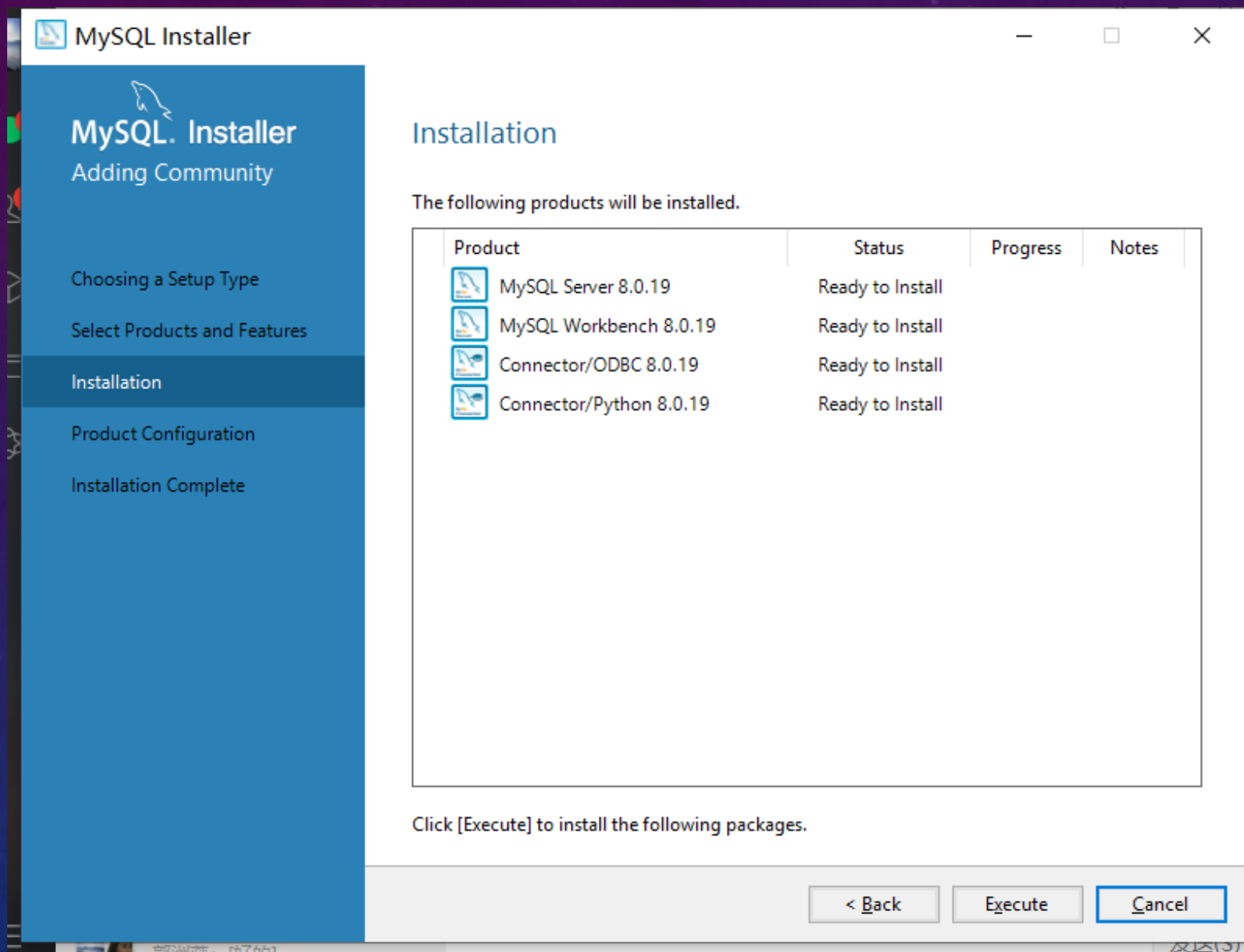
MYSQL INSTALL



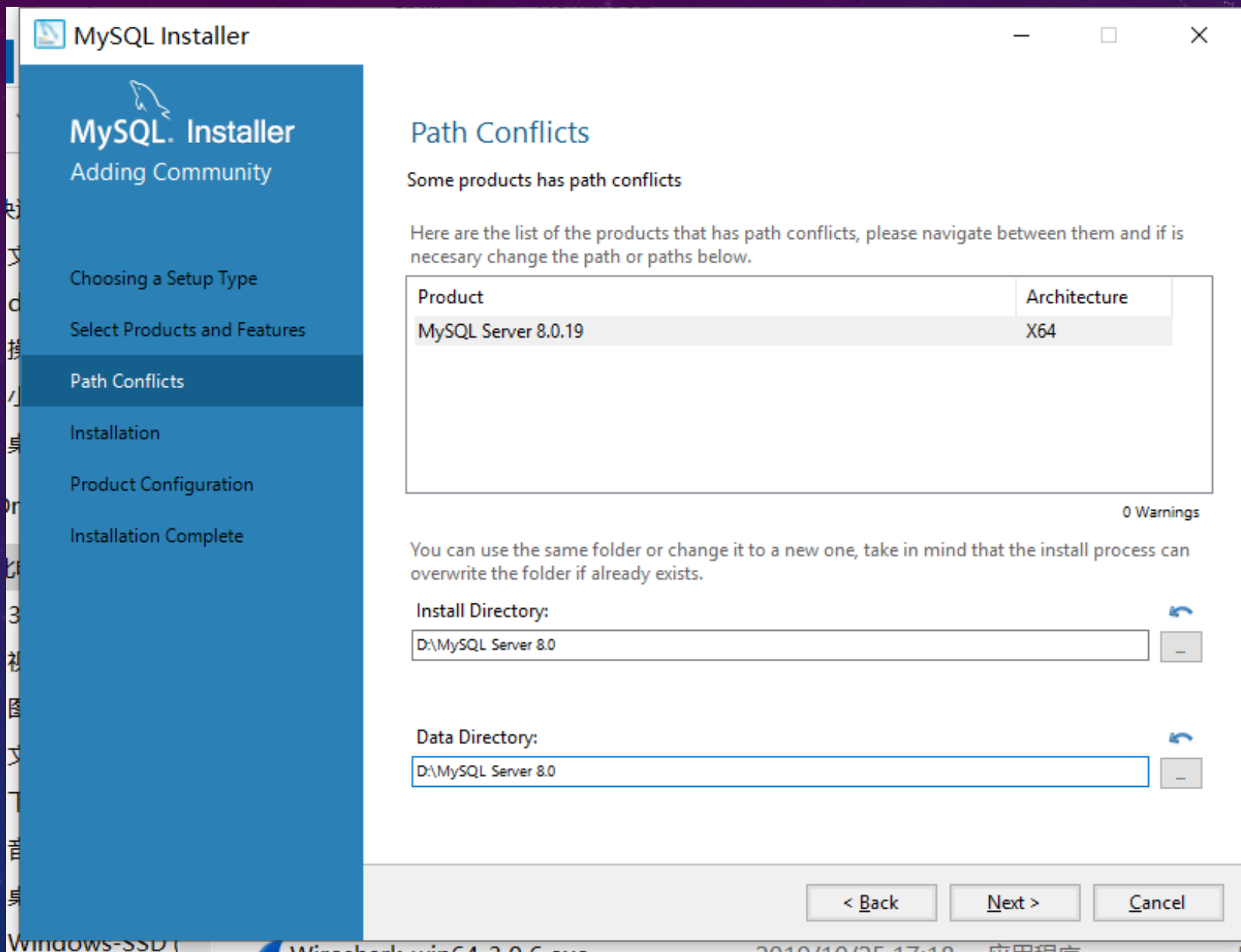
MYSQL INSTALL



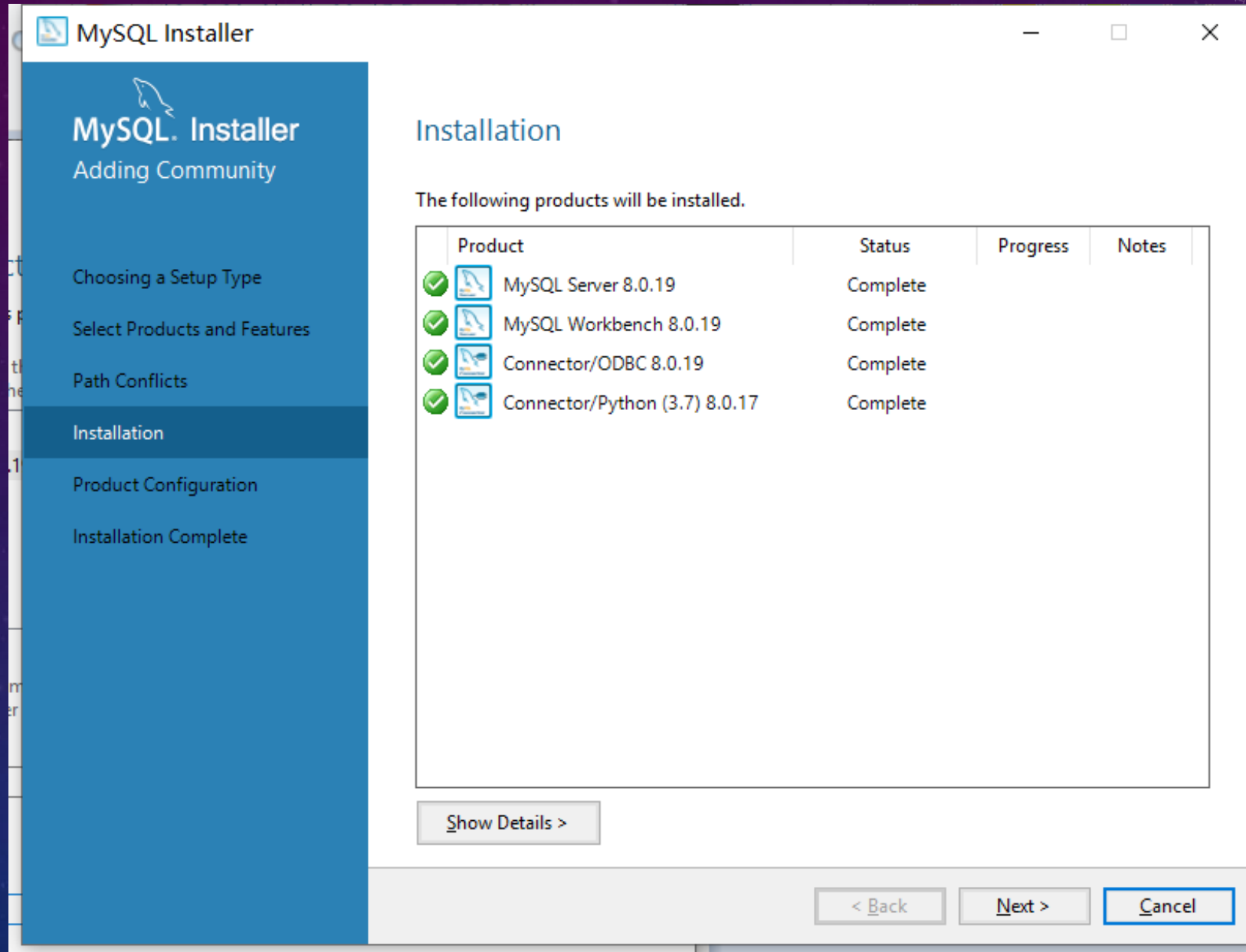
MYSQL INSTALL



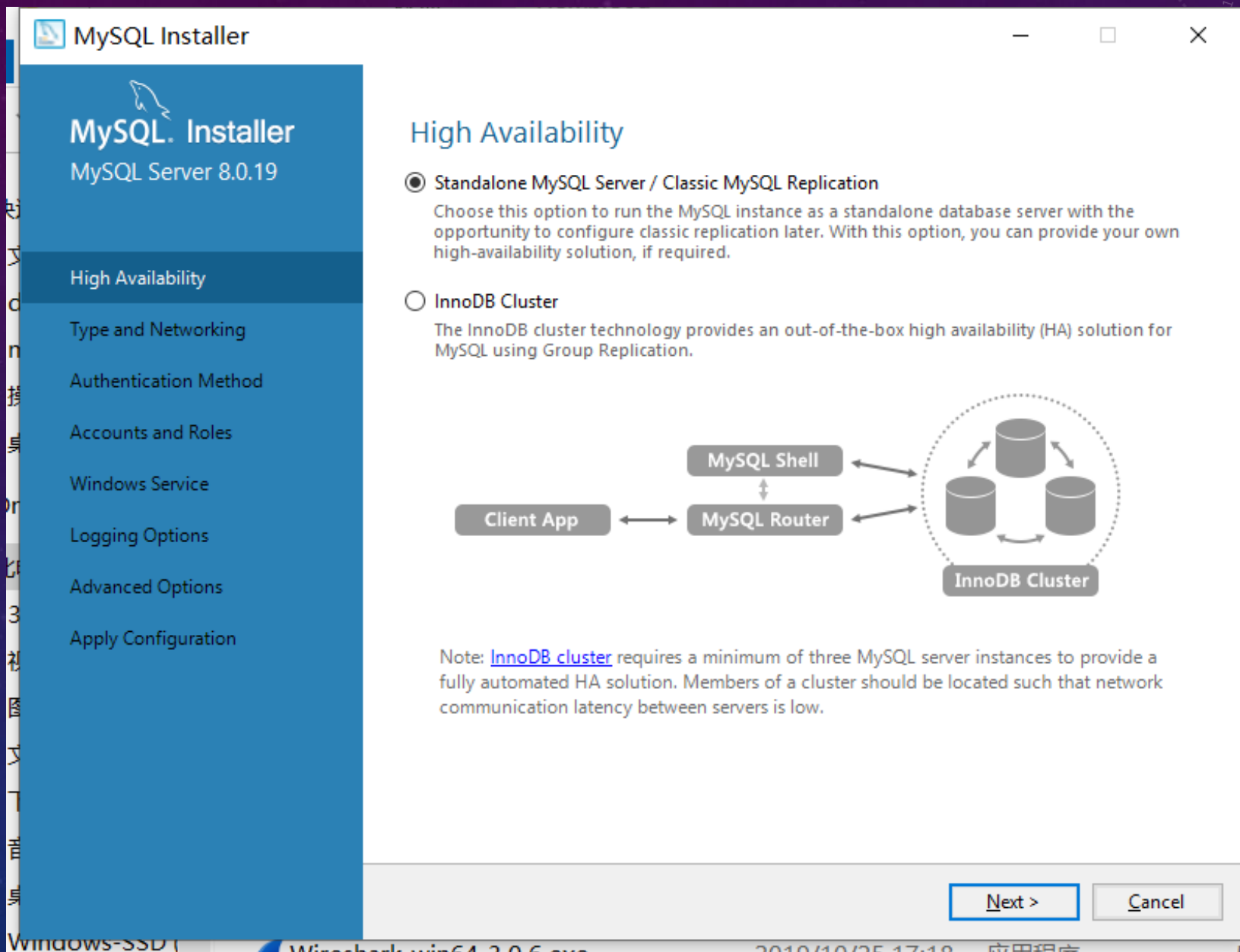
MYSQL INSTALL



MYSQL INSTALL



MYSQL INSTALL



MYSQL INSTALL

MySQL Installer
MySQL Server 5.7.17

Type and Networking

Accounts and Roles

Windows Service

Plugins and Extensions

Apply Server Configuration

Type and Networking

Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type:

Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP Port Number:

☒ Open Firewall port for network access

☐ Named Pipe Pipe Name:

☐ Shared Memory Memory Name:

Advanced Configuration

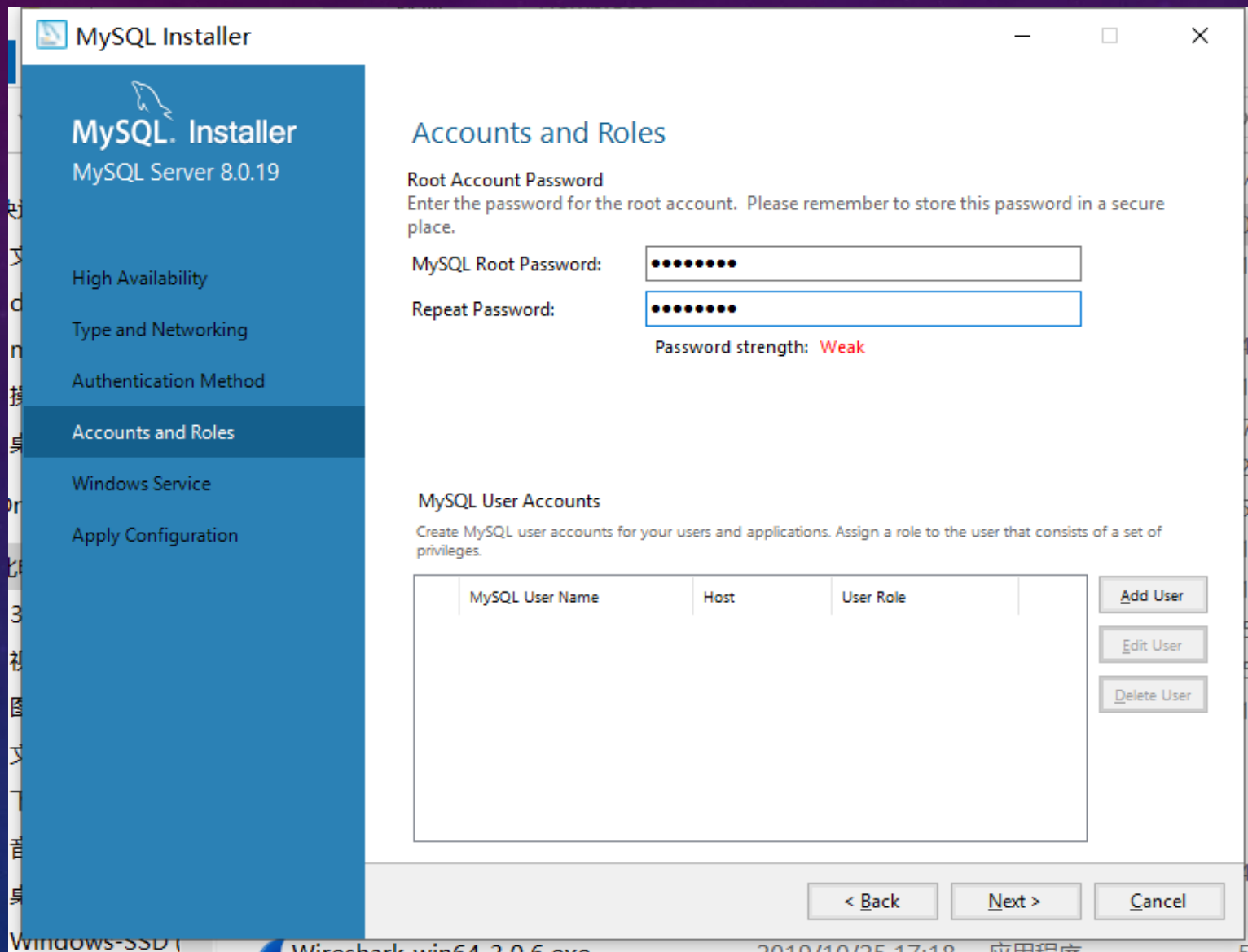
Select the checkbox below to get additional configuration page where you can set advanced options for this server instance.

☐ Show Advanced Options

<http://blog.cbdonges.com/2014/04/08/mysql-install/>

Next > Cancel

MYSQL INSTALL



The image shows the MySQL Installer window for MySQL Server 8.0.19. The left sidebar contains a list of configuration steps: High Availability, Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter a password for the root account, with fields for 'MySQL Root Password' and 'Repeat Password'. A 'Password strength' indicator shows 'Weak' in red. The second section, 'MySQL User Accounts', provides instructions to create user accounts and assign roles, accompanied by a table with columns for 'MySQL User Name', 'Host', and 'User Role'. To the right of the table are buttons for 'Add User', 'Edit User', and 'Delete User'. At the bottom of the window are navigation buttons: '< Back', 'Next >', and 'Cancel'.

MySQL Installer
MySQL Server 8.0.19

High Availability
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: **Weak**

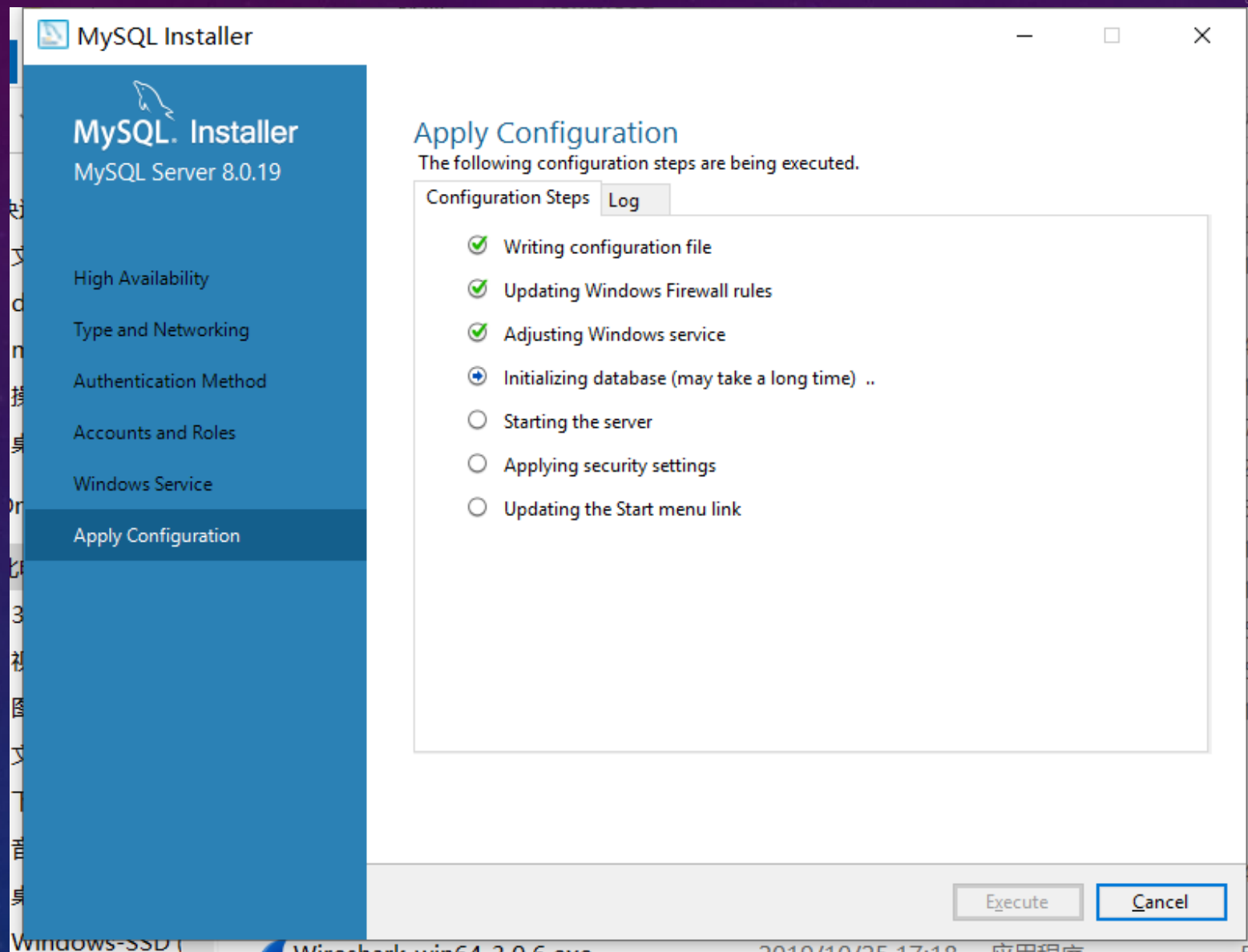
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
-----------------	------	-----------

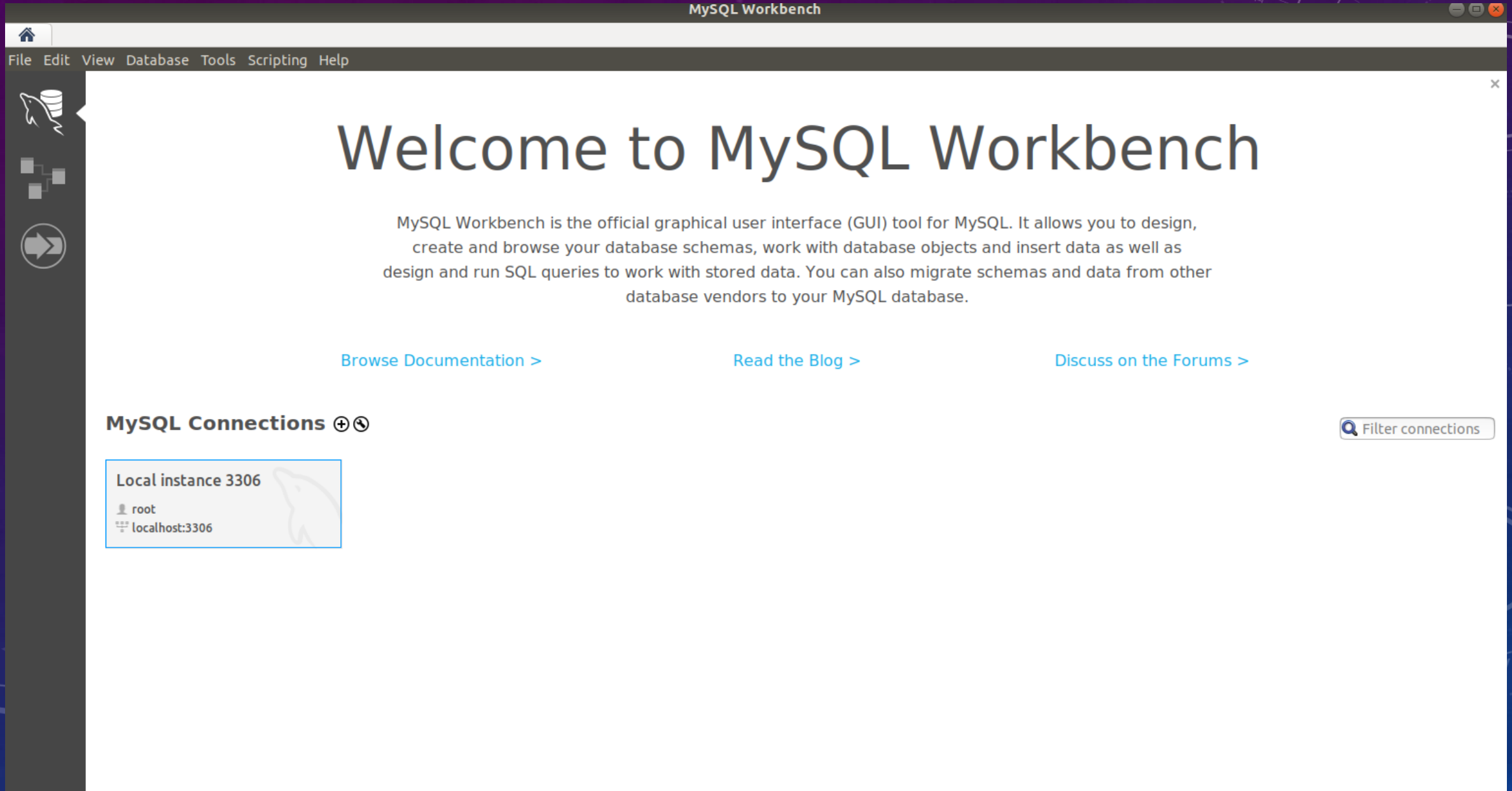
[Add User](#)
[Edit User](#)
[Delete User](#)

< Back Next > Cancel

MYSQL INSTALL



MYSQL: GUI CLIENT APPLICATION



MYSQL: GUI CLIENT APPLICATION

The screenshot displays the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar shows the SCHEMAS tree with 'sys' expanded, revealing 'sys_config' and its columns: 'variable', 'value', 'set_time', and 'set_by'. The main query editor shows a query: `select variable , value from sys.sys_config;`. The 'Result Grid' tab is active, displaying the following data:

#	variable	value
1	diagnostics.allow_i_s_tables	OFF
2	diagnostics.include_raw	OFF
3	ps_thread_trx_info.max_length	65535
4	statement_performance_analyzer.limit	100
5	statement_performance_analyzer.vi...	NULL
6	statement_truncate_len	64
*	NULL	NULL

The bottom status bar indicates 'Query Completed' and 'sys_config 1' is selected. A right-hand pane shows 'Context Help' and 'Snippets' tabs, with a note about automatic context help.

Install Python3.13 and Pycharm

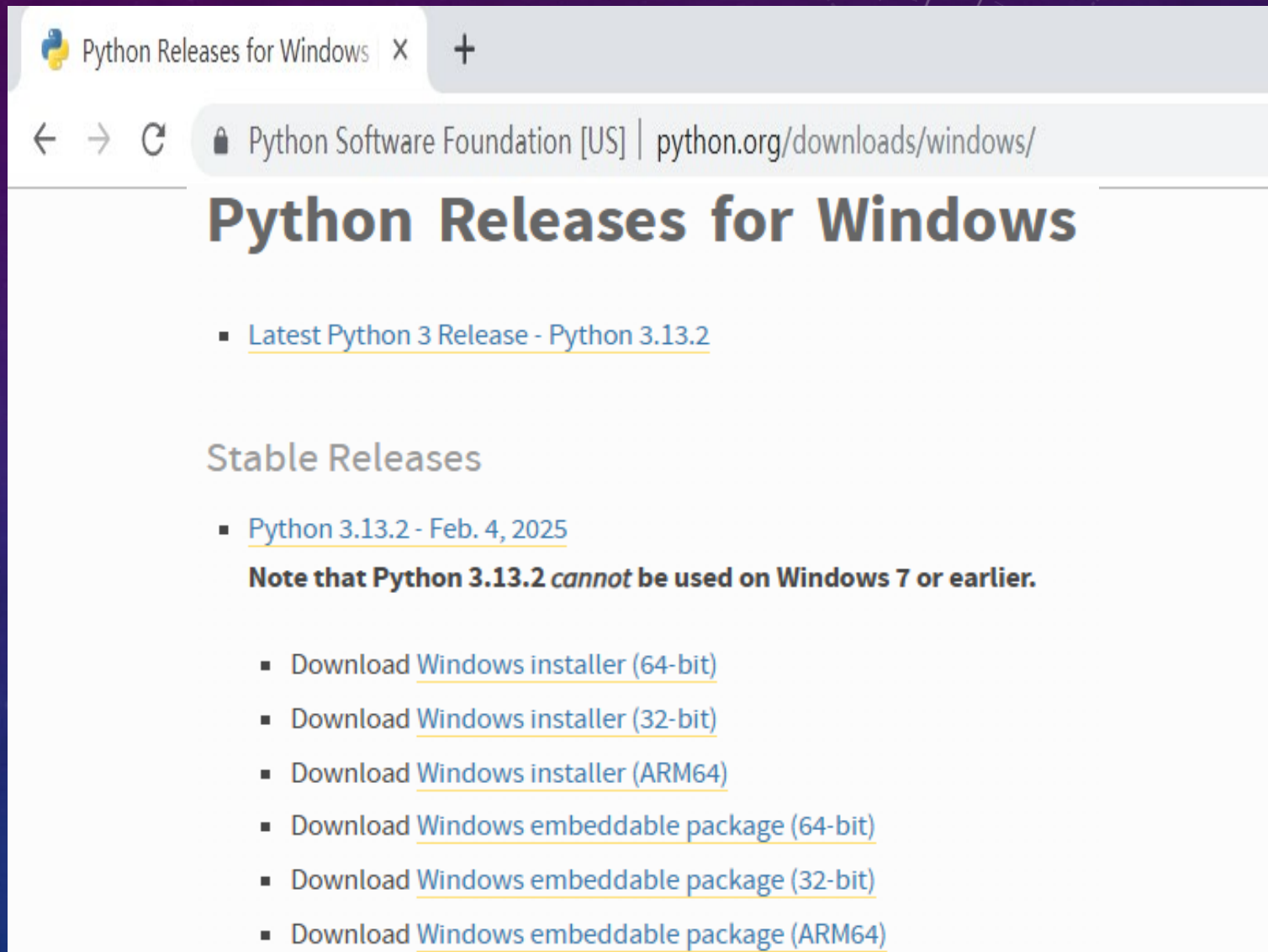
LABS PREPARE

- python

- <https://www.python.org/downloads/windows/>

- Download

- Install

A screenshot of a web browser showing the 'Python Releases for Windows' page. The browser's address bar shows the URL 'python.org/downloads/windows/'. The page title is 'Python Releases for Windows'. Below the title, there is a list of links, with the first one being 'Latest Python 3 Release - Python 3.13.2'. Underneath this, there is a section titled 'Stable Releases' with a link for 'Python 3.13.2 - Feb. 4, 2025'. A note states that Python 3.13.2 cannot be used on Windows 7 or earlier. Below the note, there are six links for downloading different versions of Python for Windows: 64-bit installer, 32-bit installer, ARM64 installer, 64-bit embeddable package, 32-bit embeddable package, and ARM64 embeddable package.

Python Releases for Windows

Python Software Foundation [US] | python.org/downloads/windows/

Python Releases for Windows

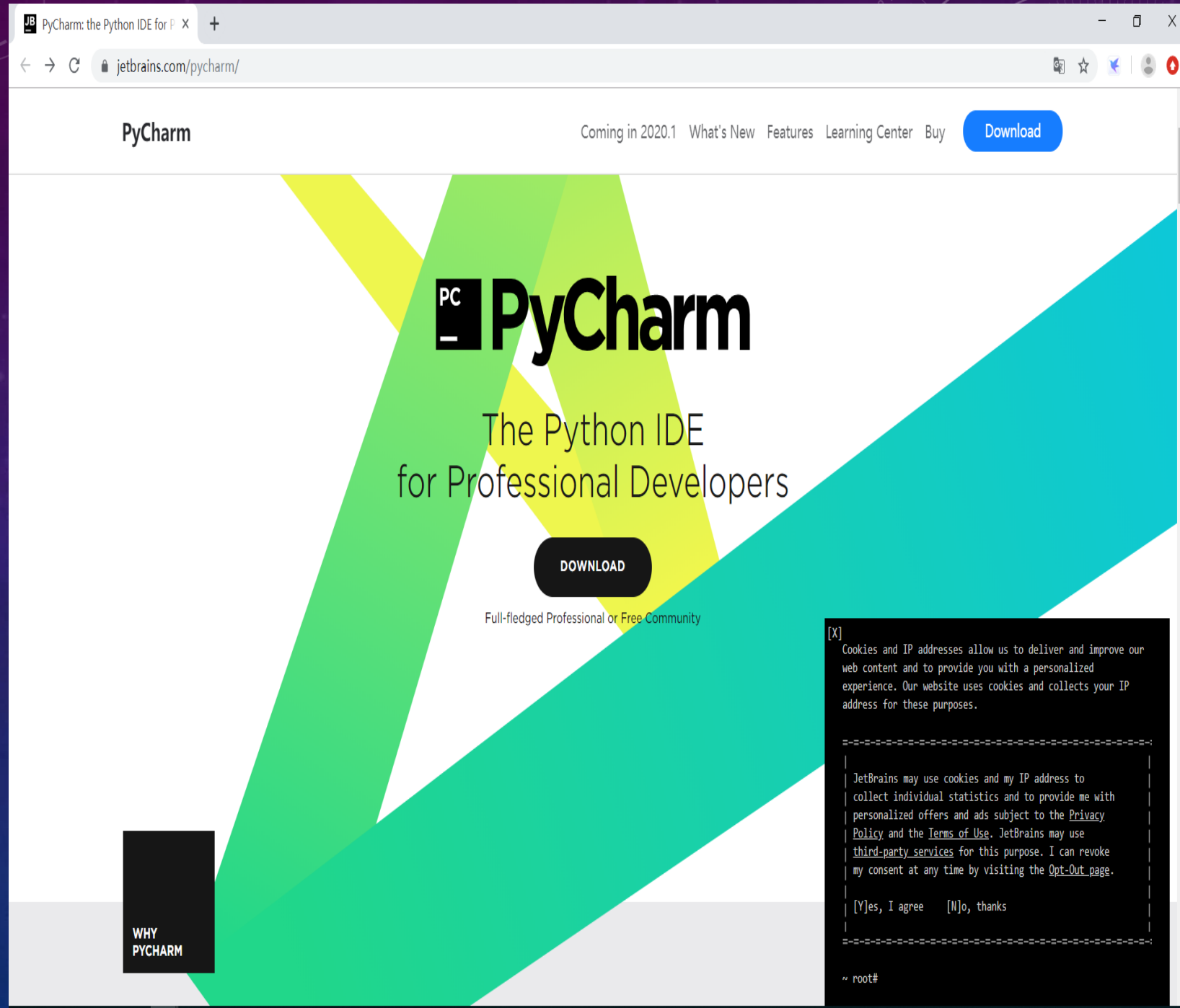
- [Latest Python 3 Release - Python 3.13.2](#)

Stable Releases

- [Python 3.13.2 - Feb. 4, 2025](#)
Note that Python 3.13.2 *cannot* be used on Windows 7 or earlier.
 - Download [Windows installer \(64-bit\)](#)
 - Download [Windows installer \(32-bit\)](#)
 - Download [Windows installer \(ARM64\)](#)
 - Download [Windows embeddable package \(64-bit\)](#)
 - Download [Windows embeddable package \(32-bit\)](#)
 - Download [Windows embeddable package \(ARM64\)](#)

LABS PREPARE

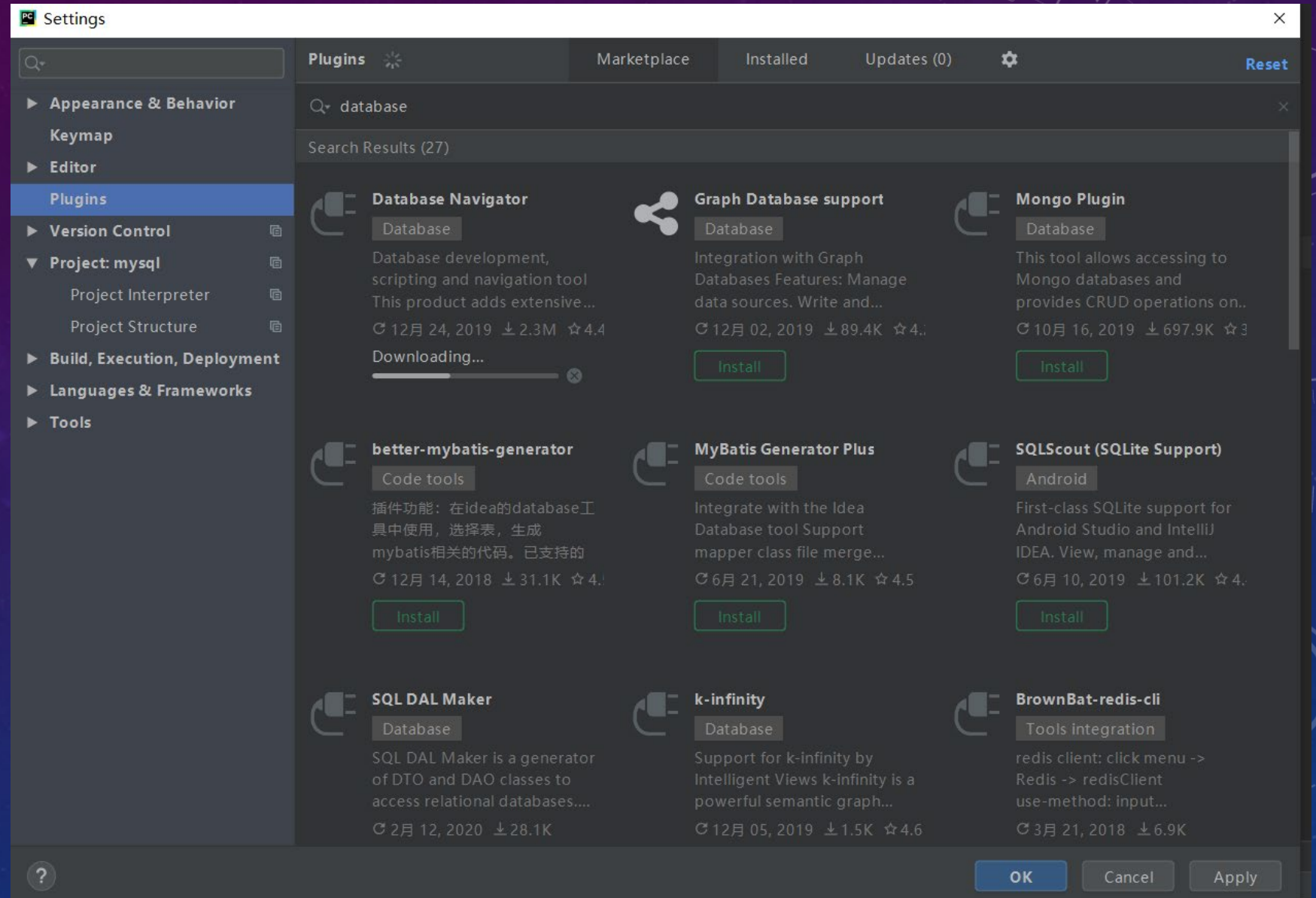
- pycharm
 - <https://www.jetbrains.com/pycharm/>
 - Download
 - Install



Connect MySQL using python

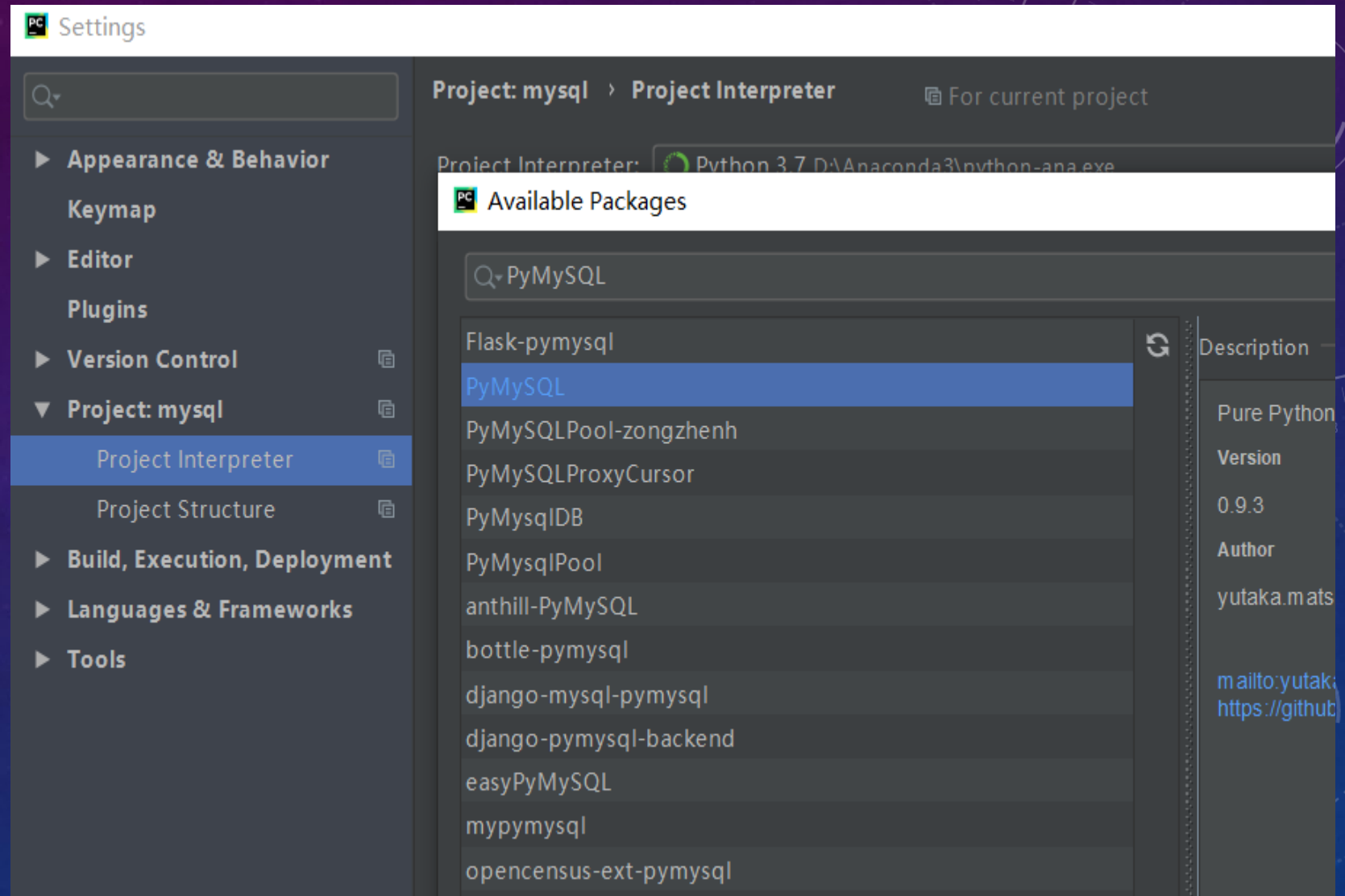
PYCHARM

- Add reference
 - PyMySQL
 - Pip install PyMySQL



PYCHARM

- Add reference
 - PyMySQL
 - Pip install PyMySQL

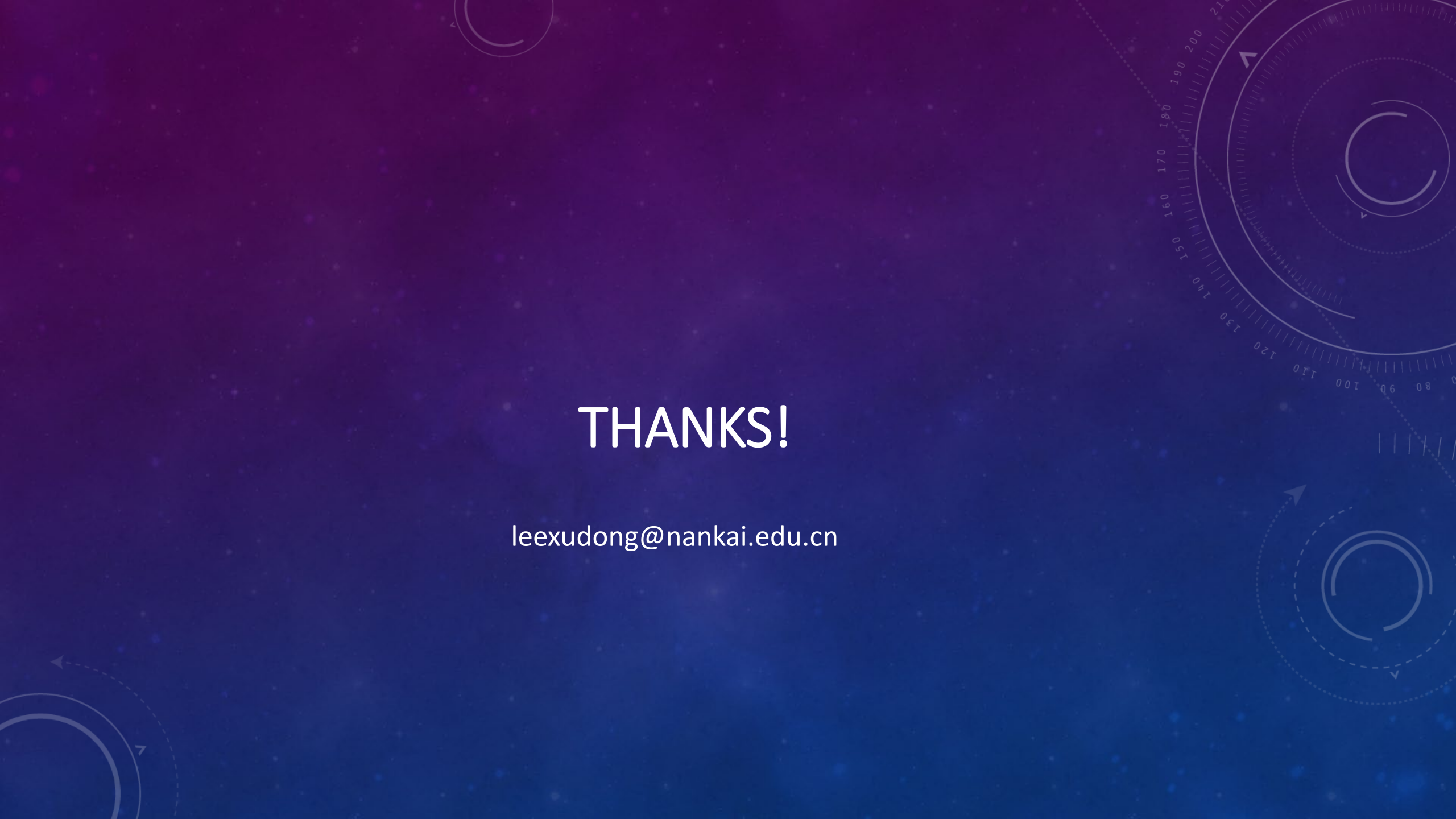


CONNECTION

```
import pymysql
conn=pymysql.connect(host='localhost',user='root',password='123',database='sys',charset='utf8')
cursor=conn.cursor()
sql = 'select version()'
cursor.execute(sql)
retdata = cursor.fetchall()
for row in retdata:
    print(row[0])
cursor.close()
conn.close()
```

Database version : 8.0.19

Process finished with exit code 0



THANKS!

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